MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

1/
TYPE 8965

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, ceramic and metal
See figure 1
Mounting position: Any
Weight: 2.5 ounces (71 grams) nominal

ABSOLUTE RATINGS:

| Parameter: | F | Ef | Eb | epy | Ec | Ik | Ib | ib | Ic | ic |
|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | MHz | V (Note 1) | Vdc | kv | Vdc | mAdc | mAdc | a | mAdc | a |
| CW Osc or Amp: | 2,500 | 6.0±5% | 1,000 | --- | -150 | 125 | --- | --- | 40 | --- |
| Anode pulsed Osc or Amp: | 3,000 | 6.0±5% | --- | 3.5 | --- | --- | --- | 3.0 | --- | 1.5 |
| Grid pulsed Osc or Amp: | 3,000 | 6.0±5% | 2,500 | --- | -150 | --- | --- | 3.0 | --- | 1.5 |
| TEST CONDITIONS: | --- | 6.0 | 1,000 | --- | Adj | --- | 100 | --- | --- | --- |

ABSOLUTE RATINGS:

| Parameter | tp | Du | Pp | Pg | tk | TE | T (Anode shank) | Cooling |
|---|---|---|---|---|---|---|---|---|
| Unit: | µs | --- | W | W (Note 2) | sec (min) | °C (Note 3) | °C (Note 3) | --- (Notes 3 & 5) |
| CW Osc or Amp: | --- | --- | 100 | 1.5 | 60 | 250 | 250 | --- |
| Anode pulsed Osc or Amp: | 6.0 | 0.0033 | 100 | 1.5 | 60 | 250 | 250 | --- |
| Grid pulsed Osc or Amp: | 6.0 | 0.0033 | 100 | 1.5 | 60 | 250 | 250 | --- |
| TEST CONDITIONS: | --- | --- | --- | --- | 300 | --- | --- | Note 6 |

GENERAL:

Qualification - Required
Compatibility testing - Required (See note 7)

1/ Formerly Y-616



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspectin, part 1 | 15 | | | | | | | |
| 1301 | Heater current | - | | 0.65 | II | If | 0.90 | 1.05 | A |
| 1261 | Electrode voltage (1) (grid) | - | | 0.65 | II | Ec | - 4.5 | -9.5 | Vdc |
| 1266 | Total grid current | - | | 0.65 | II | Ic | --- | -8.0 | µAdc |
| 1211 | Insulation of electrodes | - | Eb = Ek = 0; Ec = -500 Vdc | 0.65 | II | Rgk | 50 | --- | Meg |
| 1231 | Pulsing emission | - | eb = ec = etd/is = 6.0 a; tp = 3 µs (max); prr = 600 (max) | 0.65 | II | etd | --- | 200 | v |
| | Quality conformance inspection, part 2 | | | | | | | | |
| 1261 | Electrode voltage (2) (grid) | 8 | Ec/Ib = 1.0 mAdc | --- | --- | Eco | --- | -30.0 | Vdc |
| 1331 | Direct interelectrode capacitance (grounded cathode connection) | - | Fixture per Drawing 158-JAN | --- | --- | Cin<br>Cgp<br>Cout | 6.60<br>1.65<br>--- | 7.80<br>1.90<br>0.035 | pF<br>pF<br>pF |
| --- | Resonance | 9 | No voltages | --- | --- | --- | --- | --- | --- |
| --- | Power gain | 4 | F = 1,100 ±50 MHz; Ebb = 2,200 Vdc; Ecc = -45 Vdc; tp = 3 µs (min); Du = 0.002 (min); pd = 400 w (peak) | --- | --- | G | 6 | --- | dB |
| | Quality conformance inspection, part 3 | | | | | | | | |
| --- | Life test | 10 | Group C; Ef = 6.0 V; filament standby; t = 500 hours | --- | --- | --- | --- | --- | --- |
| --- | Life-test end points | 10 | | --- | --- | Δib<br>ΔCgp | ---<br>--- | 25<br>0.15 | %<br>pF |
| 1031 | Variable-frequency vibration | 11,12 | F = 55 to 500 to 55 Hz; Accel = 10 G peak (min); Ebb = 300 Vdc Rp = 10,000 ohms; Ec/Ib = 10 mAdc | --- | --- | Ep | --- | 250 | mVac |
| 1002 | Barometric pressure, reduced | 11,14 | Pressure = 35 mmHg (max); voltage = 1,800 Vac; TA = 30°±10°C | --- | --- | --- | --- | --- | --- |
| --- | Torque | 11,13 | No voltages | --- | --- | --- | --- | --- | --- |
| 1042 | Shock, specified pulse | 11 | Condition A; no voltages; mixture per Drawing 280-JAN | --- | --- | --- | --- | --- | --- |
| --- | Torque and shock test end point: | | | | | | | | |
| 1266 | Total grid current | - | | --- | --- | Ic | --- | - 10 | µAdc |



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NOTES:

1. The transit-time heating effect of the cathode may have to be compensated for by a reduction in the heater voltage after dynamic operation of the tube has started. This back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for any particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to 500 MHz.

2. The maximum instantaneous peak grid voltage for CW ratings shall be within the range of +30 and -400 volts; for pulse service within the range of +250 to -750 volts.

3. Sufficient conduction, convection, or forced-air cooling shall be provided for the anode shank and all seals to limit the maximum temperature to the specified value under all operating conditions. Where emphasis is placed on long and reliable life, lower temperatures should be maintained.

4. Test shall be conducted in a power amplifier cavity as shown on figure 2. Driving power is defined as the net power delivered to the amplifier cavity input terminals and the reflected power shall be subtracted from the incident power to obtain the net driving power. The output tuning shall be adjusted for maximum power output.

5. With an anode dissipation of 100 watts and with an incoming air temperature of 25°C maximum at sea level a minimum airflow of 12.5 cfm shall be directed across the anode cooler, using the cowl as shown in Drawing 157-JAN.

6. In all electrical tests involving application of heater voltage, conduction,convection, and/or forced-air cooling may be used to assure operation within the maximum rated temperature limits.

7. In addition to the requirements specified herein, compatibility testing shall be required in the AN/URN-3 and AN/GRN-9 equipments. A sample of 20 tubes shall be supplied to Naval Air Test Center (NESTED), Patuxent River, Maryland 20670, on a noninterference basis, to be tested for adequate gain and tuning capability and good tracking, in both low and high band heads. Also, of the 20 tubes supplied, a sample of four tubes shall be selected for life testing. Each tube shall achieve a minimum of 500 hours operation. In the event of no failures, the sample shall be considered as acceptable when a total of 1,600 hours has been accumulated.

8. The electrode voltage (2) grid test may be performed alternately in the following manner:

   Eb = 600 Vdc; Ec/Ib = 1.0 mAdc; and the limit for Eco shall be -18.0 Vdc maximum.

9. Grid-anode resonance. Test in cavity per Drawing 278-JAN. Cavity shall resonate at 1,354 ±2.0 MHz with tuning slug per Drawing 277-JAN at TA = 25° ±5°C.

   Grid-cathode resonance. Test in cavity per Drawing 282-JAN. Cavity shall resonate at 1,719 ±2.0 MHz with tuning slug per Drawing 277-JAN at TA = 25° ±5°C.

   When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode capacitance, the tube under test shall be represented by a point within a parallelogram whose four corners are located by the following points:

| Points | Capacitance (pF) | | Frequency (MHz) | |
|---|---|---|---|---|
| | C-gp | C-gk | F-gp | F-gk |
| 1 | 1.65 | 6.60 | 2,035 | 1,740 |
| 2 | 1.65 | 6.60 | 2,075 | 1,770 |
| 3 | 1.90 | 7.80 | 1,940 | 1,705 |
| 4 | 1.90 | 7.80 | 1,980 | 1,735 |

10. At zero hours, establish the drive conditions necessary to obtain ib = 2.0a (peak anode current) with Eb = 1,000 Vdc and a bias voltage Ec = -40 Vdc. The pulse width (tp) of the modulator shall be 2 µs minimum and the duty (Du) shall be 0.0025 maximum. With the drive level determined at zero hours, check the anode current at end of life and measure Cgp. Δib and ΔCgp may not exceed the specified limits.



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11. This test will be performed quarterly, with sampling as follows:

| | | |
|---|---|---|
| $n_1 = 4$ | $c_1 = 0$ | Where $c_2$ represents the total allowable failures for the 1st and 2nd samples combined. |
| $n_2 = 4$ | $c_2 = 1$; | |

All tests shall be considered as nondestructive except in case of failure, and separate samples may be used for each test at the option of the manufacturer. In case of failure after double sampling, the failing test shall become a QCI, part 2, AQL 6.5, inspection level S3 test, for three consecutive successful submissions, at which time the testing may revert to the QCI, part 3 basis.

12. The tube shall be mounted in a socket in accordance with Drawing 276-JAN and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within ±20 percent of the reference acceleration of 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximate logarithmic progression, and shall require 4 minutes minimum, 6 minutes maximum, to traverse the range. Each tube shall be vibrated for 30 minutes in each position X and Z, except that if the cumulative result of tests on 50 or more tubes of a construction show that more than 75 percent of the tubes have higher output voltages in one position, subsequent measurements need be taken only in the position giving the higher reading. The voltages specified herein shall be applied to the tube during vibration. The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of vibration shall be measured with a suitable device. This device shall have an appropriate voltage range and shall have the ability to measure with an error of less than 10 percent the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz. The value of the vibration output Ep shall not exceed the limit specified herein at any point in the sweep-frequency range during the last complete cycle of vibration.

13. The torque test procedure shall be as follows:

(a) A torque of 15 inch-pounds, shall be applied between the anode contact surface and the cathode without perceptiable shock.
(b) A torque of 40 inch-pounds shall be applied between the anode contact surface and the grid without perceptible shock.

14. The applied voltage shall be 60 Hz, between the anode and grid, No other voltages shall be applied. There shall be no evidence of failure as indicated by arc-over.

15. All tests listed under QCI, part 1, shall be performed at the conclusion of the holding period.

User activities:
Navy - AS, OS, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N033)



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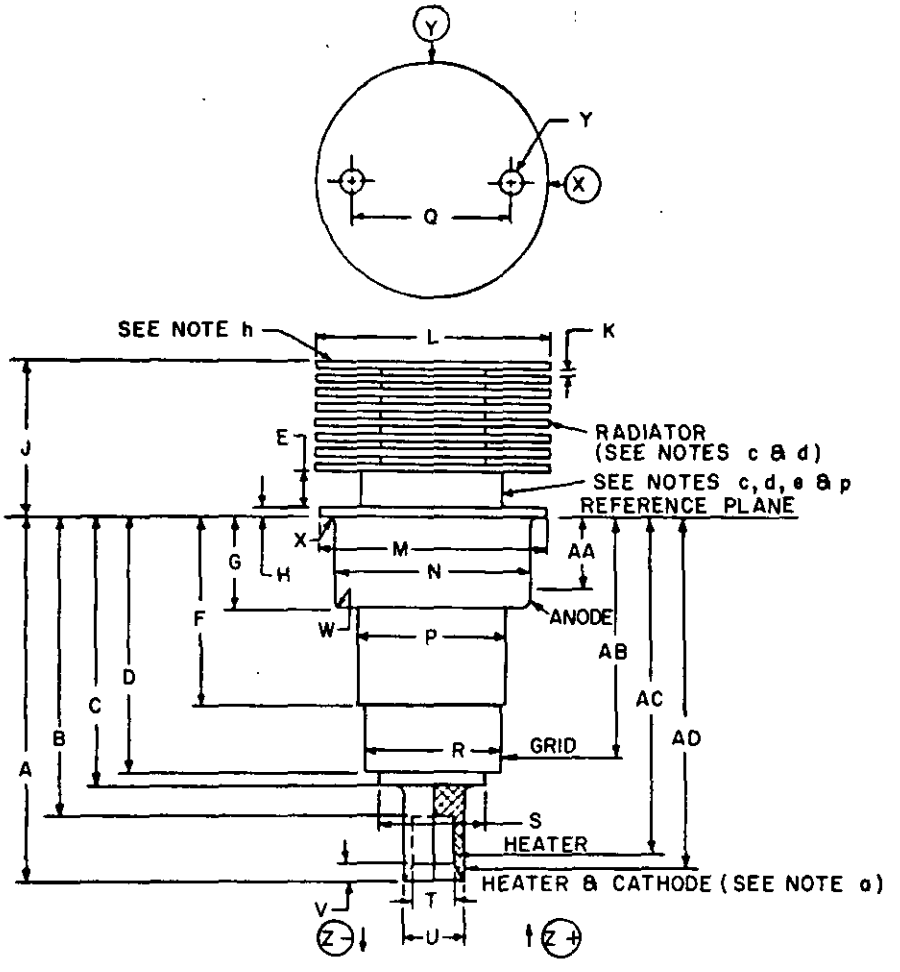



FIGURE 1. Outline drawing of electron tube type 8965.



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| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | | Notes |
|---|---|---|---|
| | Minimum | Maximum | |
| Quality conformance inspection, part 2 | | | |
| A | 1.840 (46.74) | 1.900 (48.26) | |
| B | | 1.400 (35.56) | |
| C | | 1.345 (34.16) | |
| D | 1.335 (33.91) | 1.370 (34.80) | |
| F | 0.910 (23.11) | 0.940 (23.88) | |
| G | 0.462 (11.73) | 0.477 (12.12) | |
| J | 0.766 (19.46) | 0.826 (20.98) | |
| N | 1.025 (26.04) | 1.035 (26.29) | c, k, m |
| R | 0.655 (16.64) | 0.665 (16.89) | c, k, m |
| T | 0.213 (5.41) | 0.223 (5.66) | c, g, k, m |
| U | 0.315 (8.00) | 0.325 (8.26) | c, g, k, m |
| Quality conformance inspection, part 3 | | | 11 |
| E | 0.125 (3.18) | 0.185 (4.70) | |
| H | | 0.040 (1.02) | |
| K | 0.025 (0.64) | 0.046 (1.17) | j |
| L | 1.234 (31.34) | 1.264 (32.11) | |
| M | 1.180 (29.97) | 1.195 (30.35) | |
| P | 0.752 (19.10) | 0.792 (20.12) | n |
| Q | 0.650 (16.51) | 0.650 (21.59) | |
| S | | 0.545 (13.84) | |
| V | | 0.086 (2.18) | |
| W | | 0.100 (2.54) RAD | |
| X | | 0.035 (0.89) RAD | |
| Y | 0.105 (2.67) | 0.145 (3.68) | f |
| Electrode contact areas | | | b |
| AA | 0.110 (2.79) | 0.436 (17.07) | |
| AB | 0.850 (21.59) | 1.250 (31.75) | |
| AC | 1.380 (35.05) | 1.780 (45.21) | |
| AD | 1.410 (35.81) | 1.820 (46.23) | |

NOTES:

a. Insulation material between heater and heater-cathode shall be securely affixed.
b. Dimensions in electrode contact area table are for socket design purposes and are not intended for inspection purposes.
c. Silver plated 30 MSI minimum.
d. Plating not required over radiator and radiator support of copper, aluminum, or approved equivalent.
e. This surface shall be used for measurement of anode shank temperature.
f. Holes for tube extractor through top fin only.
g. Inner edge of heater and outer edge of cathode rf connection shall be free from burrs and sharp edges.
h. This fin shall withstand a 6-inch (152.40 mm) drop test without loosening and without distortion as judged by ability to maintain dimension K.
j. Distortion of fins permissible provided distance between adjacent fins at any point on circumference meets tolerances for dimension K.
k. Eccentricity of contact surfaces shall be gaged from center line of reference shall be as follows:

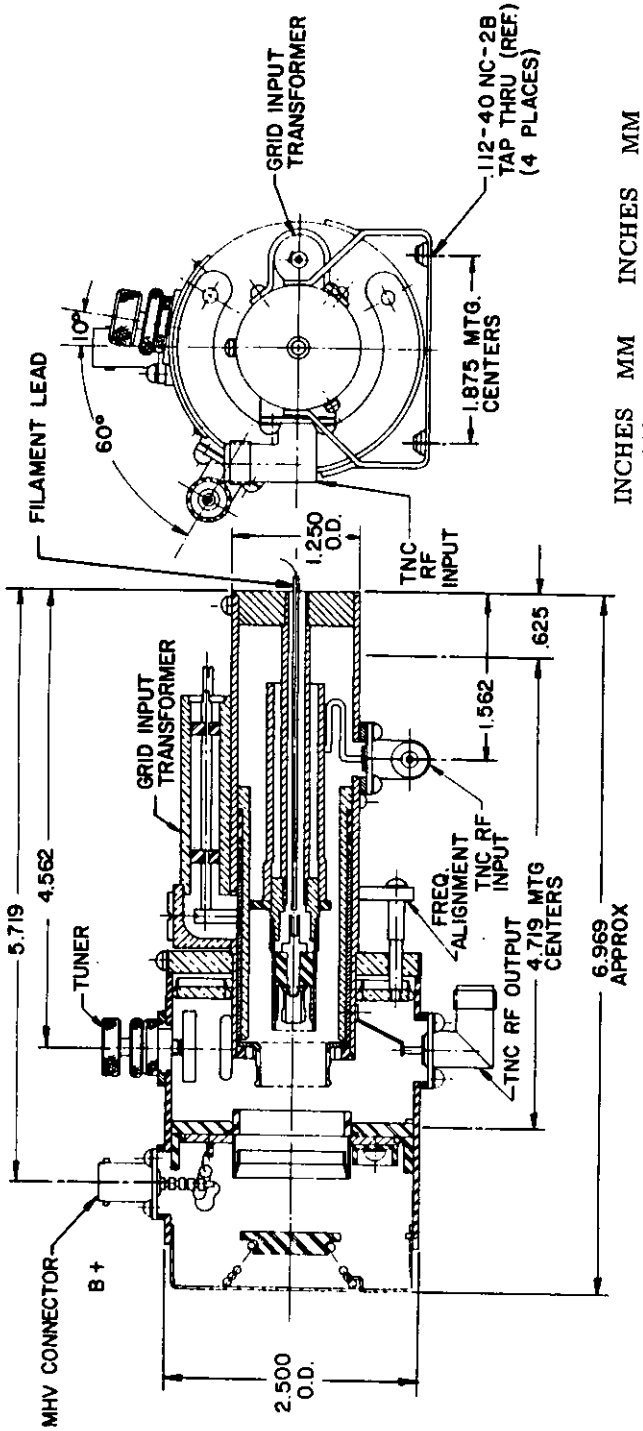
| Contact surface | TIR maximum | Reference |
|---|---|---|
| Anode | 0.020 | Cathode |
| Grid | 0.020 | Cathode |
| Heater | 0.012 | Cathode |

m. Diameters N, R, T, and U shall apply throughout entire contact areas as defined by dimensions AA, AB, AC, and AD respectively.
n. This surface shall not be used for clamping or locating.

FIGURE 1. Outline drawing of electron tube type 8965 - Continued.



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| INCHES | MM |
|---|---|
| .112 | 2.84 |
| .625 | 15.88 |
| 1.250 | 31.75 |
| 1.562 | 39.67 |
| 1.875 | 47.63 |

| INCHES | MM |
|---|---|
| 2.500 | 63.50 |
| 4.562 | 115.87 |
| 4.719 | 119.86 |
| 5.719 | 145.26 |
| 6.969 | 177.01 |

NOTE: All dimensions are for reference only.

FIGURE 2. Pulse amplifier cavity.



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